

main.c	Run	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 int main() 4 { 5 int r[50], n, head, size, dir; 6 int i, j, temp, total = 0; 7 printf("Enter number of disk requests: "); 8 scanf("%d", &n); 9 printf("Enter disk request queue: "); 10 for(i = 0; i < n; i++) 11 scanf("%d", &r[i]); 12 printf("Enter initial head position: "); 13 scanf("%d", &head); 14 printf("Enter disk size: "); 15 scanf("%d", &size); 16 printf("Enter direction (1-Right, 0-Left): "); 17 scanf("%d", &dir); 18 for(i = 0; i < n - 1; i++) 19 for(j = i + 1; j < n; j++) 20 if(r[i] > r[j]) 21 { 22 temp = r[i]; 23 r[i] = r[j]; 24 r[j] = temp; 25 } 26 printf("\nSeek Sequence:\n%d", head); 27 if(dir == 1) 28 { 29 for(i = 0; i < n; i++) 30 if(r[i] >= head) 31 { 32 total += abs(head - r[i]); 33 head = r[i]; 34 printf(" -> %d", head); 35 } 36 total += abs(head - (size - 1)); 37 head = size - 1; 38 printf(" -> %d", head); 39 for(i = n - 1; i >= 0; i--) 40 if(r[i] < head) 41 { 42 total += abs(head - r[i]); 43 head = r[i]; 44 printf(" -> %d", head); 45 } 46 } 47 else 48 { 49 for(i = n - 1; i >= 0; i--) 50 if(r[i] <= head) 51 { 52 total += abs(head - r[i]); 53 head = r[i]; 54 printf(" -> %d", head); 55 } 56 total += abs(head - 0); 57 head = 0; 58 printf(" -> %d", head); 59 }</pre>		Enter number of disk requests: 6 Enter disk request queue: 20 35 70 90 10 50 Enter initial head position: 45 Enter disk size: 100 Enter direction (1-Right, 0-Left): 1 Seek Sequence: 45 -> 50 -> 70 -> 90 -> 99 -> 90 -> 70 -> 50 -> 35 -> 20 -> 10 Total Head Movement = 143 cylinders Average Head Movement = 23.83 cylinders == Code Execution Successful ==